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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I B. Avinash (192415296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B. Avinash

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

1. Evolution from Standalone computing to cloud-Based systems.

Computing has evolved through several major stages:

Key technology transitions:-

1. Standalone computing: applications and data were stored and processed on a single computer.
2. Client-Server architecture: clients accessed centralized servers for applications, databases and services.
3. Distributed computing: multiple computers worked together over networks to share processing and storage.
4. Internet and high speed networking: global connectivity enabled remote access to computing resources.
5. Virtualization: A single physical server could run multiple virtual machines improving resource utilization.
6. Service oriented architecture and APIs: applications were divided into reusable services that could communicate with each other.

Evanta

Difference between Type 1 and Type 2 Hypervisors

<u>Feature</u>	<u>Type 1 Hypervisor</u>	<u>Type 2 Hypervisor</u>
-Also called	Bare-metal hypervisor	hosted hypervisor
Runs on	Directly on physical hardware	on top of a host operating system
Performance	Higher	comparatively lower
Security	more isolated	Depends partly on host OS
usage	Data centers and cloud platforms	desktop testing and development
examples	VMware ESXi, microsoft hyperv, xen	Oracle virtual box, VMware workstation

Example: A cloud data center may use a type 1 hypervisor to run hundreds of virtual machines on a single physical server.

3 Role of APIs in cloud service integration

An API (application program interface) is a set of rules and protocols that allows different software applications and cloud services to communicate.

Role in cloud integration.

APIs allow applications to:

- Access cloud storage and databases
- Start, stop and manage virtual machines
- Transfer data between different cloud services
- Integrate applications from different cloud providers
- Automate cloud operations

APIs and interoperability:

APIs support interoperability by providing standardized methods for different systems to exchange data and services.

Example:

An e-commerce application can use:

- A payment API for payment processing
- A cloud storage API for storing product images
- A shipping API for delivery tracking.

Resource Pooling in Cloud Computing.

resource pooling is a cloud computing concept in which computing resources are shared among multiple users through a common pool.

Resources include:

- CPU processing power
- memory
- storage
- network bandwidth
- virtual machines.

The cloud provider dynamically allocates these resources to users according to their requirements.

Benefits for multiple users:

Better resource utilization: unused resources can be allocated to other users.

Cost efficiency: users pay only for the resources they consume.

Scalability: Resources can be increased or decreased as needed.

Multi-tenancy: multiple users can share the same physical infrastructure.

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Rapid Elasticity in cloud Computing.

Rapid elasticity is the ability of cloud resources to quickly expand or shrink according to workload requirements.

Cloud System can automatically:

- Add more virtual machines during high demand.
- Increase CPU and memory resources.
- Remove unused resources when demand decreases.

Support for dynamic workloads:

For example, an online shopping website may experience:

- normal traffic: 5 servers are sufficient.
- During a festival sale: traffic increases significantly so the cloud automatically adds more servers.
- After the sale: extra servers are removed.

This enables organizations to handle sudden changes in workload without purchasing permanent hardware.

Key benefit: Rapid elasticity provides flexibility, scalability and cost efficiency by matching computing resources with real-time demand.